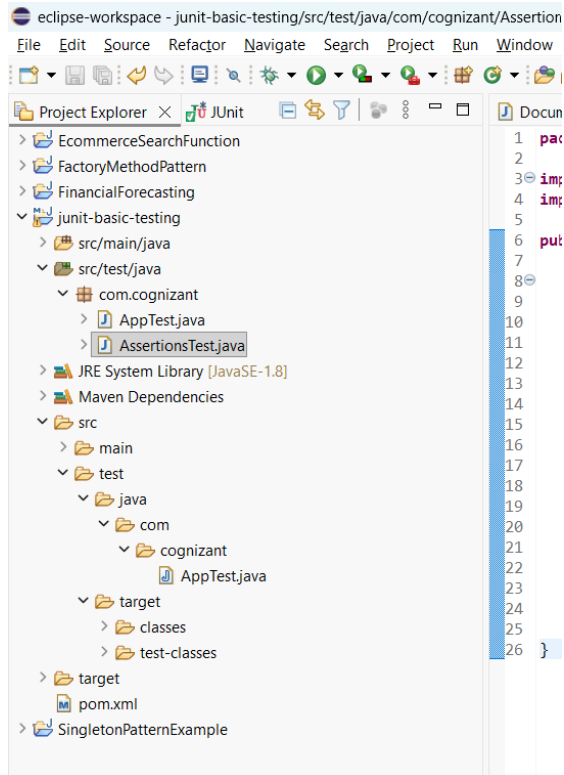


JUnit Testing Exercises

Exercise 3: Assertions in JUnit

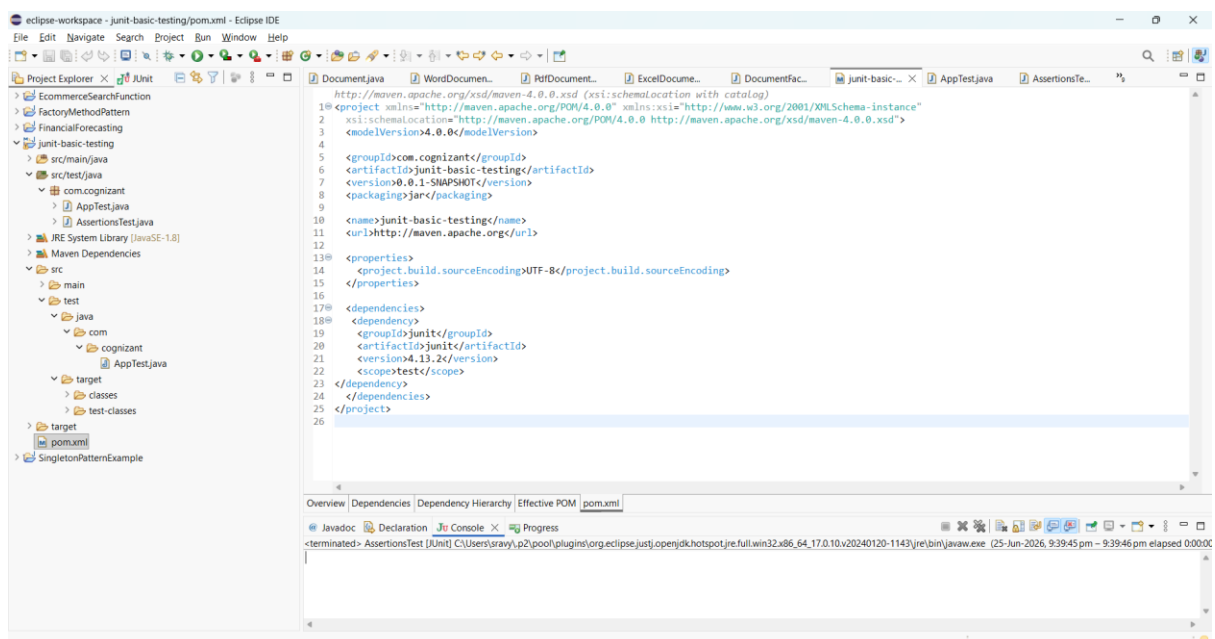
Step 1: Open the Existing Maven Project

Opened the existing Maven project junit-basic-testing in Eclipse IDE.



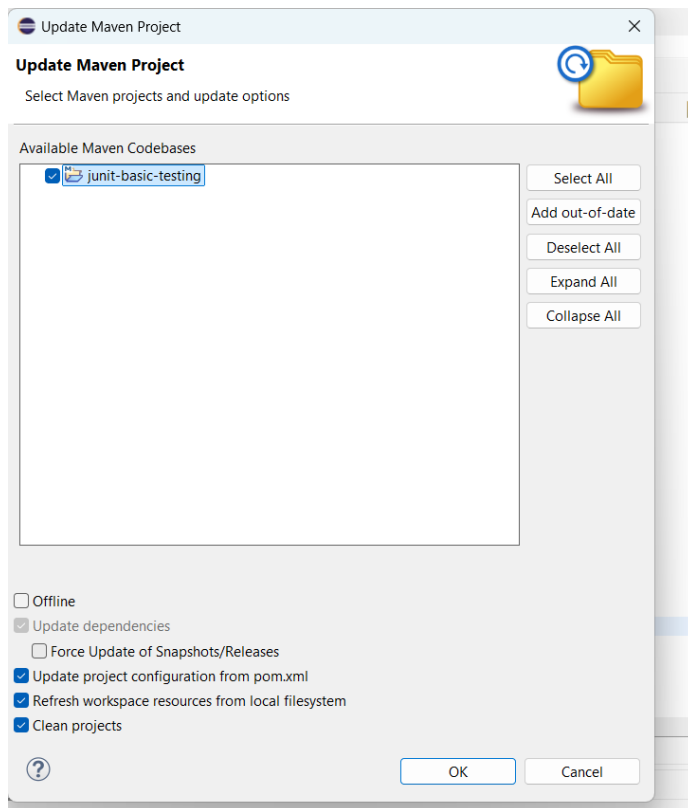
Step 2: Verify the JUnit Dependency

Opened the pom.xml file and verified that the JUnit 4.13.2 dependency was added successfully.



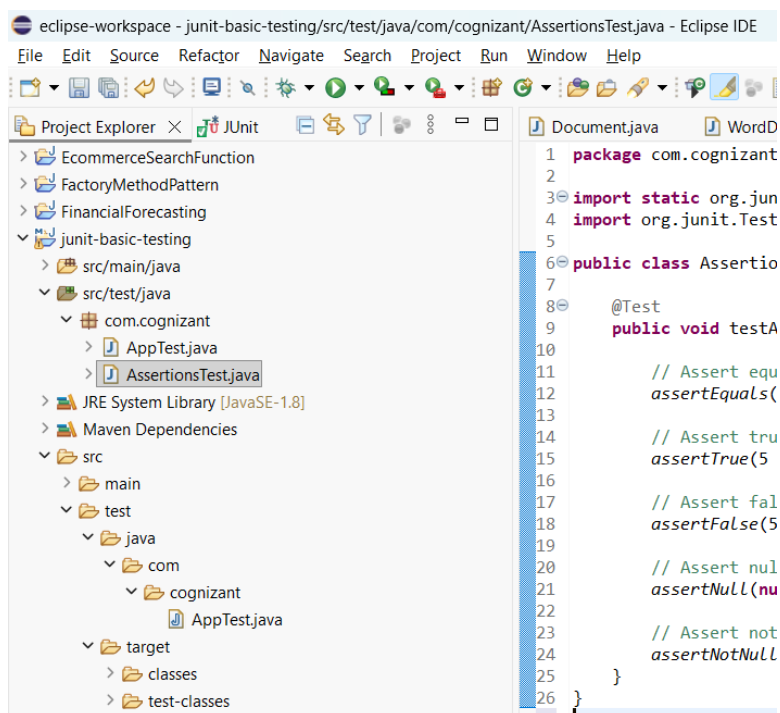
Step 3: Update the Maven Project

Updated the Maven project using Maven → Update Project to download and configure the JUnit dependency.



Step 4: Create the AssertionsTest Class

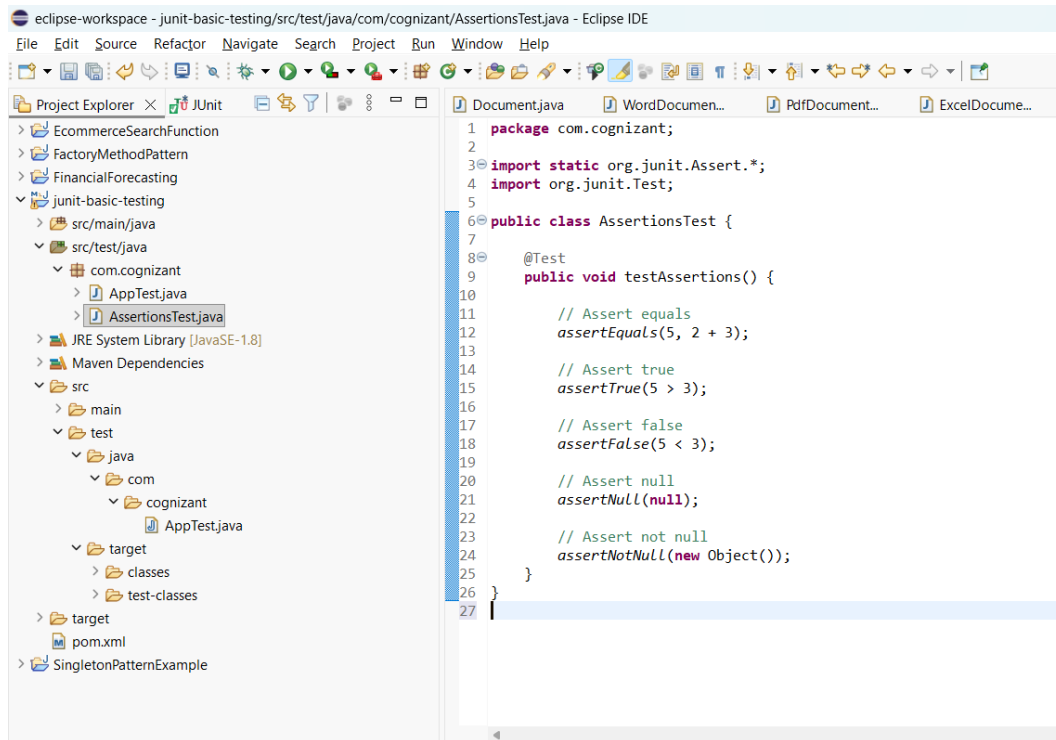
Created a new Java class named AssertionsTest inside the src/test/java/com.cognizant package.



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Step 5: Write the JUnit Assertion Program

Implemented different JUnit assertion methods to verify expected results.



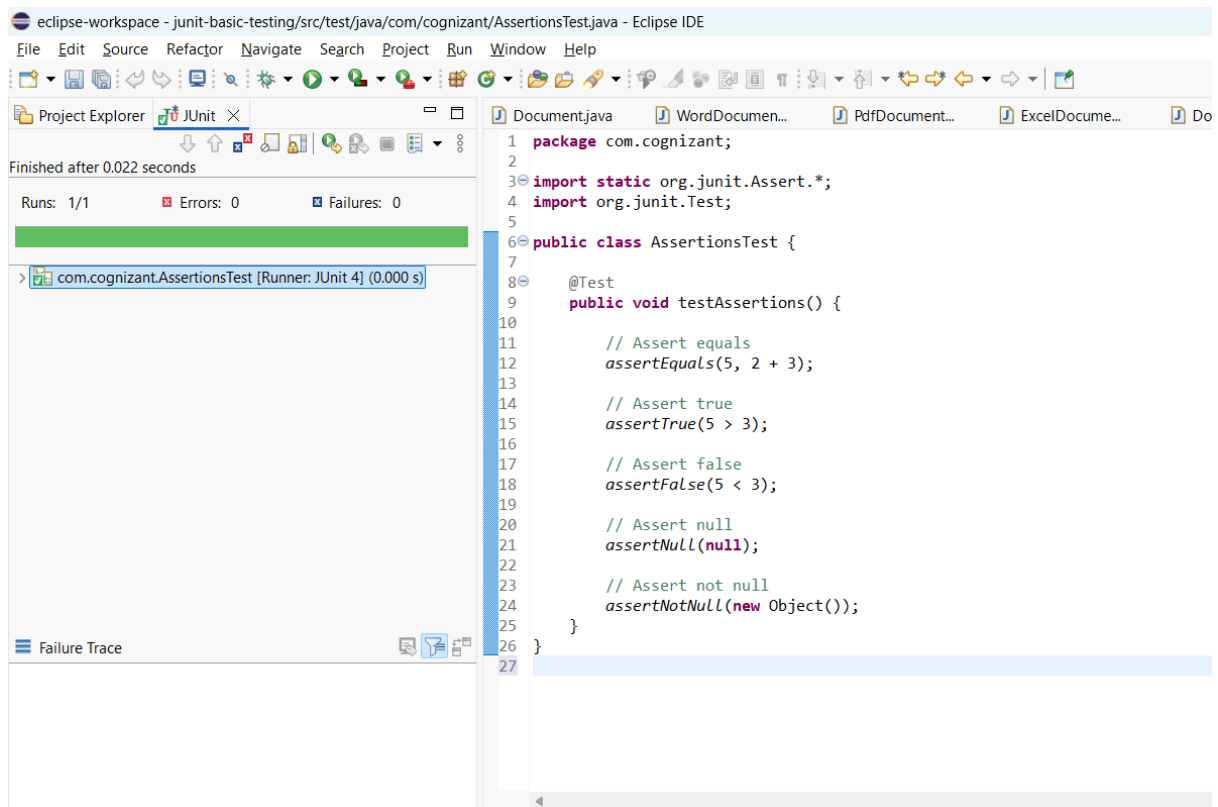
```
eclipse-workspace - junit-basic-testing/src/test/java/com/cognizant/AssertionsTest.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help

Project Explorer
> EcommerceSearchFunction
> FactoryMethodPattern
> FinancialForecasting
> junit-basic-testing
  > src/main/java
  > src/test/java
    > com.cognizant
      > AppTest.java
      > AssertionsTest.java
    > JRE System Library [JavaSE-1.8]
    > Maven Dependencies
  > src
    > main
    > test
      > java
        > com
          > cognizant
            AppTest.java
        > target
          > classes
          > test-classes
    > target
    > pom.xml
  > SingletonPatternExample

Document.java WordDocument... PdfDocument... ExcelDocume...
1 package com.cognizant;
2
3 import static org.junit.Assert.*;
4 import org.junit.Test;
5
6 public class AssertionsTest {
7
8     @Test
9     public void testAssertions() {
10
11         // Assert equals
12         assertEquals(5, 2 + 3);
13
14         // Assert true
15         assertTrue(5 > 3);
16
17         // Assert false
18         assertFalse(5 < 3);
19
20         // Assert null
21         assertNull(null);
22
23         // Assert not null
24         assertNotNull(new Object());
25     }
26 }
27
```

Step 6: Execute the JUnit Test

Executed the AssertionsTest class using Run As → JUnit Test.



```
eclipse-workspace - junit-basic-testing/src/test/java/com/cognizant/AssertionsTest.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help

Project Explorer
> EcommerceSearchFunction
> FactoryMethodPattern
> FinancialForecasting
> junit-basic-testing
  > src/main/java
  > src/test/java
    > com.cognizant
      > AppTest.java
      > AssertionsTest.java
    > JRE System Library [JavaSE-1.8]
    > Maven Dependencies
  > src
    > main
    > test
      > java
        > com
          > cognizant
            AppTest.java
        > target
          > classes
          > test-classes
    > target
    > pom.xml
  > SingletonPatternExample

JUnit
Finished after 0.022 seconds
Runs: 1/1 Errors: 0 Failures: 0
> com.cognizant.AssertionsTest [Runner: JUnit 4] (0.000 s)

Document.java WordDocument... PdfDocument... ExcelDocume... Do
1 package com.cognizant;
2
3 import static org.junit.Assert.*;
4 import org.junit.Test;
5
6 public class AssertionsTest {
7
8     @Test
9     public void testAssertions() {
10
11         // Assert equals
12         assertEquals(5, 2 + 3);
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14         // Assert true
15         assertTrue(5 > 3);
16
17         // Assert false
18         assertFalse(5 < 3);
19
20         // Assert null
21         assertNull(null);
22
23         // Assert not null
24         assertNotNull(new Object());
25     }
26 }
27
```

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Output

Runs : 1/1

Errors : 0

Failures : 0

All assertions executed successfully.

Result

The JUnit assertion methods were implemented successfully. The test executed without any errors or failures, confirming that all assertions passed successfully.